

Tracking the Evolution of AI, Machine Learning, and NLP Research Discourse (2000–2021) from Theory to Application

Individual Report

Course: Introduction to AI and Text Analytics
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1 Contribution and Activity

1.1 Project Overview and My Role

1 This project studied how summaries of AI, ML, and NLP research changed from 1990 to 2025, focusing on three main

2 questions: how research moved from theory to practical

2 work; how the way limitations were described changed over

2 time; and whether new terms spread through the field by be-

2 ing used more often or by being used in new ways. A fourth

2 overall question guided the use of multiple analysis methods

2 to determine whether different text analysis techniques agree

2 on when and how these changes occurred, or whether they

2 yield very different results.

3 My role included both managing the project and handling

key technical tasks. From the first meeting on 18 March,

the team came together. To manage the project, I set up

our Jira board and kept the shared documents up to date. I

collected data preparation work from all five members and

combined it into a standard data cleaning process, which I

shared on GitHub as the basis for all subsequent analyses.

Towards the end, since we were drafting the report on GDocs,

I converted all the GDocs to a L^AT_EX document on Overleaf,

organized its structure, added each member’s analysis sec-

tions, and completed the report for submission. Apart from

this I built a three part system for detecting semantic change

- terminology diffusion using tfidf, word2vec and specter2.

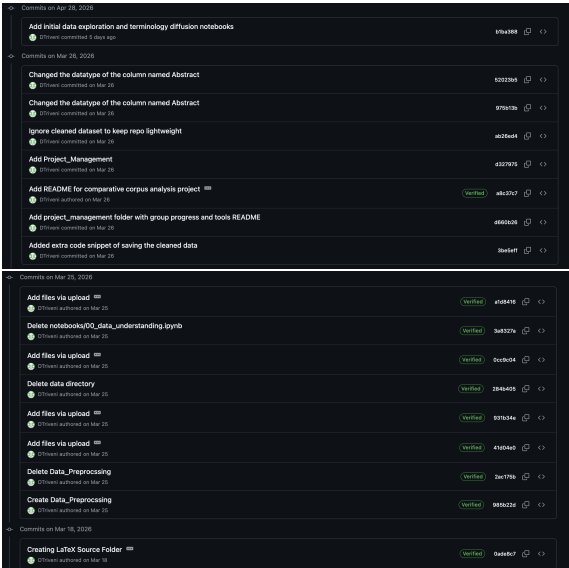


Figure 1: GitHub repository showing the shared data cleaning pipeline and version control.

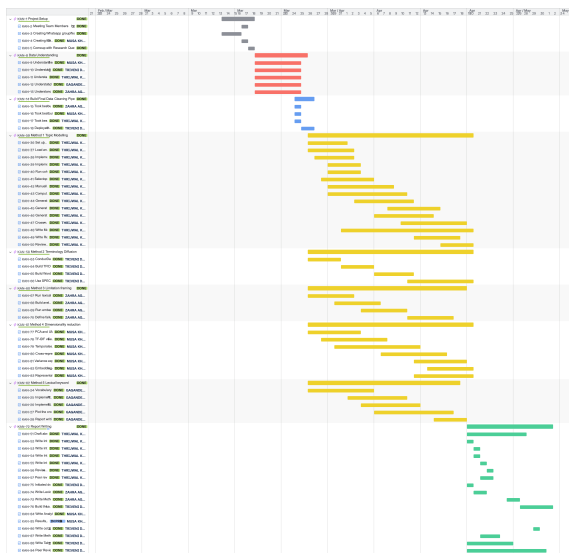


Figure 2: Team Gantt chart illustrating the coordination of five workstreams across the project timeline.

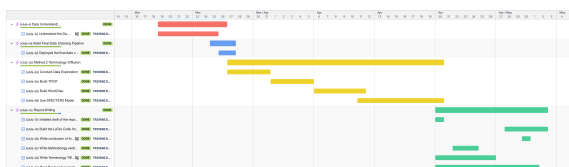


Figure 3: Individual Gantt chart detailing my tasks: project management, data cleaning pipeline, $\text{\LaTeX}$ conversion, and final report assembly.

2 Technical Understanding

2.1 My Contribution: A Three-Part System for Detecting Semantic Change

After setting up the process, the team worked together to explore the data, focusing on abstract lengths, category makeup, and time coverage. This helped decide to group data into 5-year periods, set a 50-word minimum for abstracts, and limit the variables used. Next, the team chose five methods to analyze the language: BoW, TF-IDF, Word2Vec, Sentence-BERT, and SPECTER2, each focusing on a different aspect of how language changes.

My subquestion asked whether new terms spread by simply appearing more often or by changing their meaning. The main challenge was distinguishing these signals. TF-IDF alone could not tell real changes in meaning from terms that just appeared more often without changing. I brought this up in a team meeting, which led to the idea that signals about word use and context should add to surface frequency, not replace it.

Based on this, I created a three-part system:

- **TF-IDF change** – measured how much a term was adopted on the surface;
- **Word2Vec** – measured shifts in word context by comparing models from different time periods using a special alignment on 1,436 key terms;

- **SPECTER2** – captured changes in how terms were used in full abstracts, with a method to stabilise results for small samples.

All three signals were scaled to 0–1 and a cutoff of 0.5 was set, creating four groups: terms actively spreading and those copied as templates. By showing that TF-IDF, Word2Vec, and SPECTER2 reveal distinct signals rather than simply repeating the same information, my analysis gave clear evidence for the group’s main point: no single method can entirely capture how scientific language changes over time. The consistent process I created ensured that all five subquestions used the same data, allowing fair comparisons between methods throughout the report.

3 Team Work

3.1 Team Collaboration and Work Allocation

The team worked on a set weekly schedule, meeting online via Google Meet every Monday and in person every Thursday to keep all five workstreams coordinated throughout the project. From the start, I set up a Jira board to track tasks and a shared Google Doc to record decisions, with WhatsApp used for lighter communication during the week.

Every meeting followed the same format: members shared progress, asked questions about methods, and agreed on next steps together. This was especially helpful when choosing methods, where the decision on five text representations came from open team discussion, and in making key decisions like the 5-year period grouping, category selection, and minimum abstract threshold, all recorded so everyone could work from the same analysis base.

Each team member contributed a clearly defined analytical workstream, with integration managed centrally. Table 1 summarises the roles and deliverables.

Table 1: Team member workstreams and responsibilities

Member	Roles / Workstreams
Triveni	Project Management; Jira Administration; Shared Data Cleaning pipeline; Terminology Diffusion Analysis; Report $\LaTeX$ Documentation & Finalisation; Peer Review
Musa	Project Management; EDA; PCA and temporal drift visualisation; Representation and Semantic Drift analysis; Report Documentation; Peer Review
Thrijwal	EDA; Topic Modelling; Focus Shift analysis; Report Documentation; Peer Review
Gagandeep	EDA; Lexical Analysis; Focus Shift analysis; Report Documentation; Peer Review
Zahra	EDA; Limitations Framing analysis; Report Documentation; Peer Review

All workstream outputs were integrated through the central Jira board and shared document, ensuring that the final report and analysis reflect consistent, cross-validated contributions from the whole team.

4 Conclusion

This project gave a great chance to work through the entire process of a text analytics pipeline, from collecting raw data to comparing different methods and writing the final academic report. Leading the team while also handling a major part of the analysis meant balancing coordination tasks with technical work, which I found both challenging and rewarding at the master's level.